

DVWA WEB APPLICATION SECURITY LAB REPORT

Author: Abdulmajid Bello

Course: Cybersecurity Practical Lab

Date: May 2026

Lab Title: Accessing and Configuring DVWA on Metasploitable 2 Using Kali Linux

Summary of the report.

This report documents the successful deployment of a controlled web application security lab using Kali Linux as the attacker machine and Metasploitable 2 as the target system. The primary objective was to verify network connectivity between both systems, access the Damn Vulnerable Web Application (DVWA), and configure the application security level to Low in preparation for vulnerability testing.

1. Objective

- Configure two virtual machines to communicate on a Host-Only network.
- Verify IP addressing and connectivity.
- Access the Metasploitable 2 web server from Kali Linux.
- Log in to DVWA using default credentials.
- Change the DVWA security level from High to Low.
- Prepare the environment for future ethical hacking exercises.

2. Lab Environment

- Virtualization Platform: Oracle VM VirtualBox
- Attacker Machine: Kali Linux
- Victim Machine: Metasploitable 2
- Vulnerable Application: DVWA
- Web Browser: Firefox
- Network Type: Host-Only Adapter

3. Network Configuration

- Kali Linux (eth0): 192.168.169.5
- Metasploitable 2 (eth0): 192.168.169.6

4. Methodology

4.1 Configure Virtual Network

Both virtual machines were configured to use a Host-Only Adapter in VirtualBox.

4.2 Verify IP Addresses

The ifconfig command was used on both systems.

4.3 Test Connectivity

From Kali Linux:

ping 192.168.169.6

4.4 Access the Web Server

Using Firefox on Kali Linux:

http://192.168.169.6

4.5 Launch DVWA

The DVWA link was selected from the Metasploitable homepage.

4.6 Authenticate to DVWA

Username: admin

Password: password

4.7 Change Security Level

DVWA Security was changed from High to Low.

5. Results

- Host-Only network configured successfully.
- Connectivity verified using ping.
- Metasploitable web server accessed successfully.
- DVWA login successful.
- Security level changed to Low.

The environment is now ready for:

- SQL Injection
- Cross-Site Scripting (XSS)
- Command Injection
- File Inclusion
- Brute Force Testing

6. Evidence Collected

- Host-Only network configuration screenshot
- Kali ifconfig output
- Metasploitable ifconfig output

- Ping results
- Metasploitable homepage
- DVWA login page
- DVWA dashboard
- Security level set to Low

7. Security Considerations

DVWA is intentionally vulnerable and should only be used in an isolated lab environment for educational purposes.

8. Challenges Encountered

No major technical issues were encountered.

9. Final Note

This lab successfully established a practical environment for learning web application security testing. Kali Linux and Metasploitable 2 communicated over a Host-Only network, and DVWA was configured at a low security level for future penetration testing exercises.

10. Recommendations

- Perform Nmap scanning
- Use Burp Suite
- Practice SQL Injection and XSS
- Capture traffic with Wireshark
- Document findings and remediation

11. References

- <https://www.kali.org/>
- <https://sourceforge.net/projects/metasploitable/>
- <https://github.com/digininja/DVWA>
- <https://www.virtualbox.org/>